big_house

Drone-derived roof measurement

Property	big_house
Flight date	2026-07-11 to 2026-07-12
Pipeline version	roofkit 0.1.0 (bdecd8a)
Freeze commit	4fe6859
Report date	2026-07-20

Area is scale-confirmed by an independent tape length, not validated against a measured area (none exists for this property). See the Validation page.

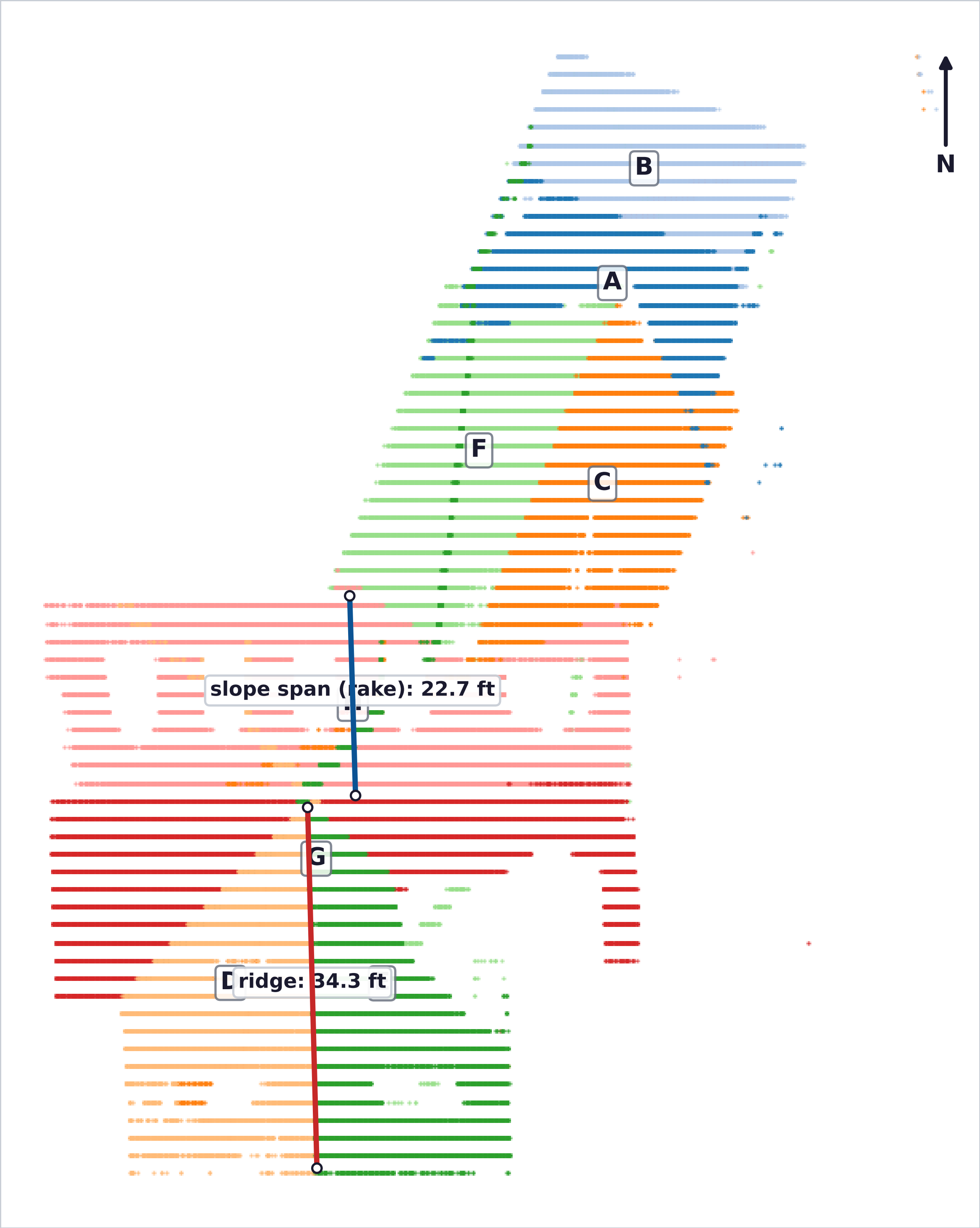
SUMMARY

3,559.3 ft²
total roof area

8
roof facets

34° (8:12)
predominant pitch

Top-down orthographic. Facet labels A-H (smallest area first). Only the two cloud-derived, tape-validated lines are dimensioned.

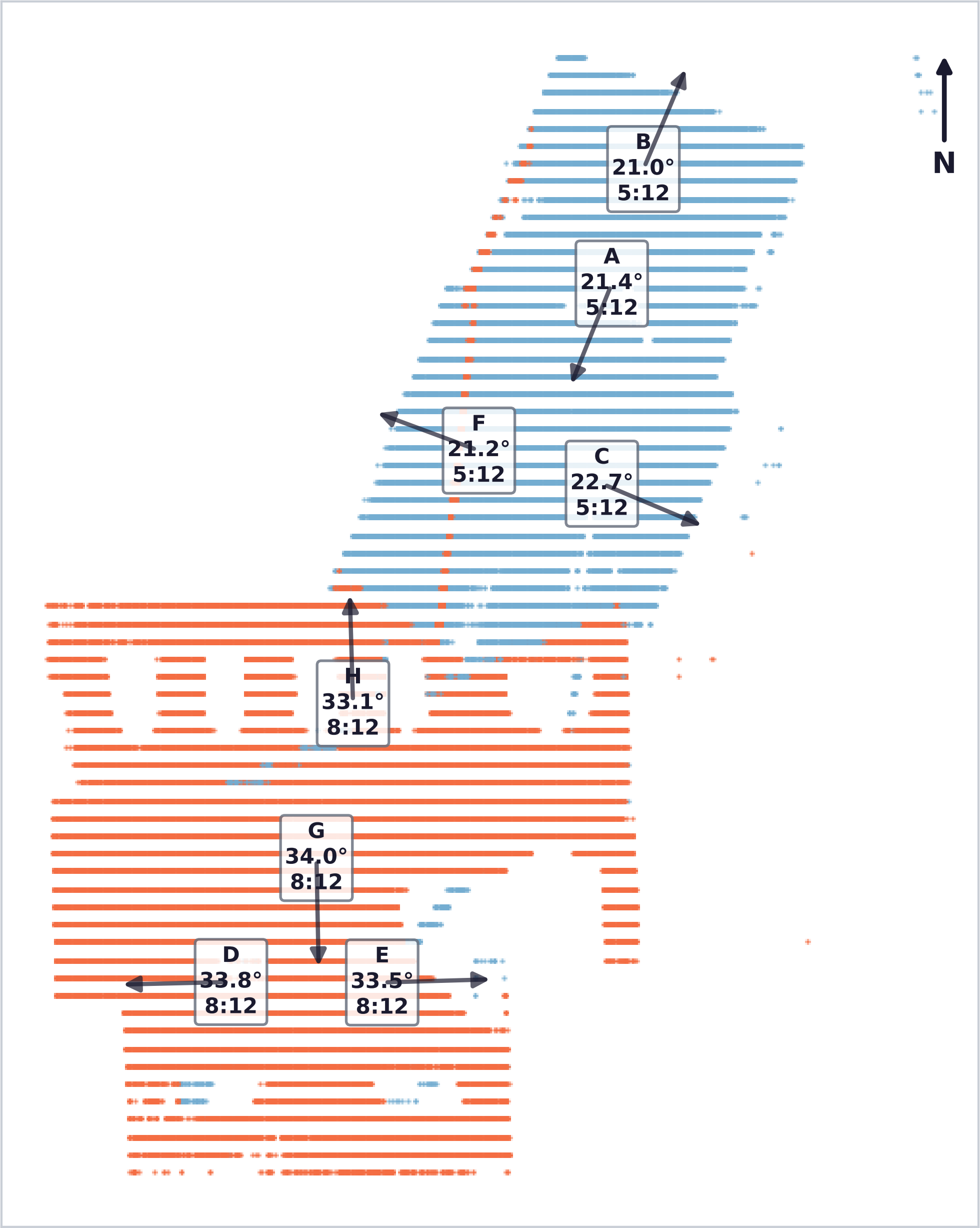


Dimensioned lines (the only two with a validated length; every other edge is an outline, not a measured edge):

Primary length:r6,7 (ridge): cloud-derived 34.33 ft; tape 34.33 ft.

Fallback lines:r1,3-e1 (slope span (rake)): cloud-derived 22.68 ft; tape 22.25 ft, 22.50 ft with gutter.

Facets colored by pitch band; arrow points downslope. Pitch is scale-independent.

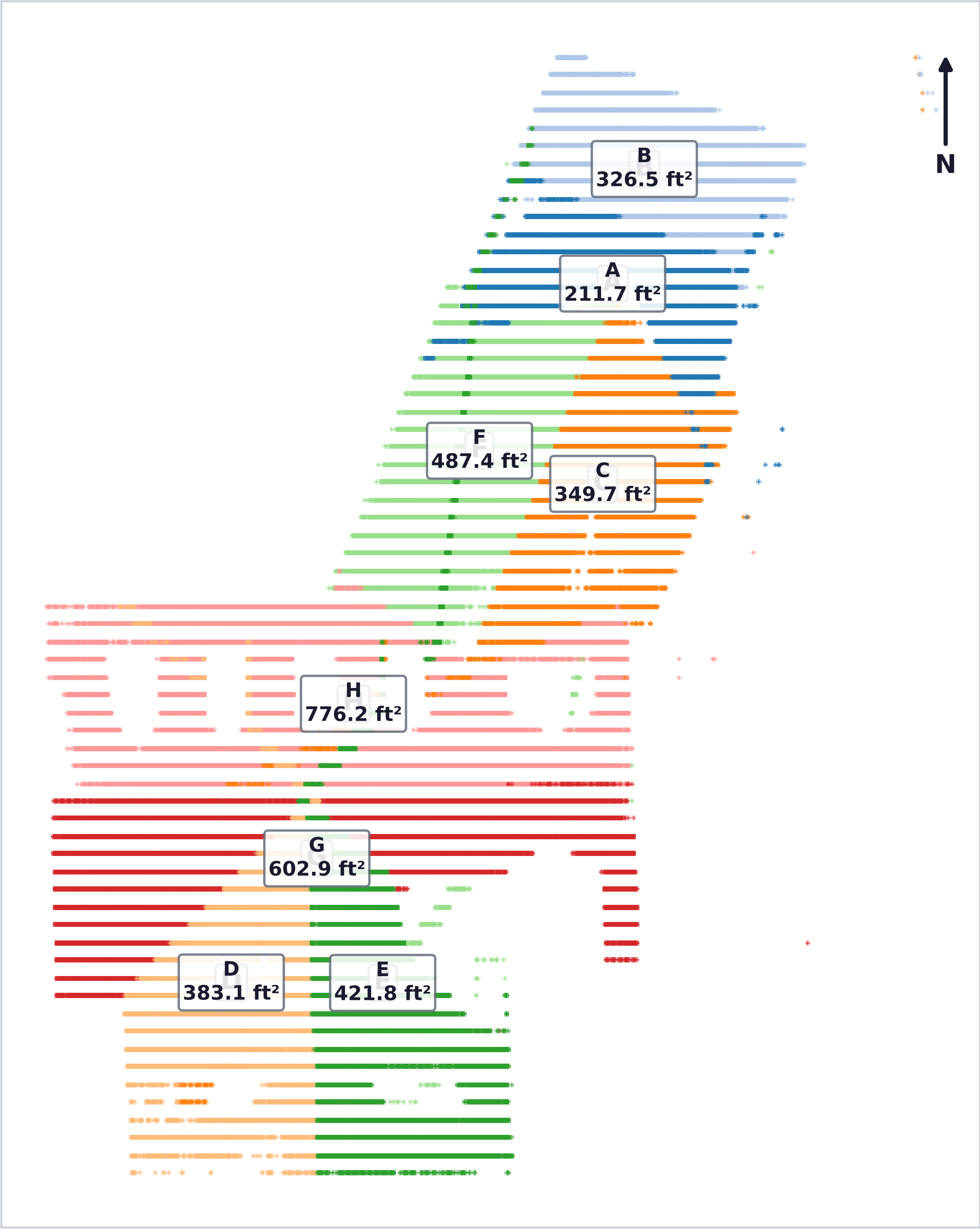


Pitch bands:

- Shallow (10-26 deg)
- Steep (26-38 deg)

The x:12 value under each facet is a rounding of the degree measurement to the nearest inch of rise per 12 inches of run, not an independent measurement.

Per-facet slope area (ft²) labeled in place. Slope area is what shingles cover, not footprint.



Total slope area: 3,559.3 ft²

Total carries a dormer caveat (see facet table and validation page); it is not a clean sum.

One row per facet, labels A-H by area. x:12 is a rounding of the degree column.

Label	Seg#	Area ft²	Pitch °	x:12	Azimuth °	Dormer	Eave	Panel
A	5	211.7	21.4	5:12	202	clean	flag	-
B	4	326.5	21.0	5:12	23	clean	-	-
C	2	349.7	22.7	5:12	113	susp*	flag	-
D	6	383.1	33.8	8:12	269	susp	flag	-
E	7	421.8	33.5	8:12	88	susp	-	-
F	0	487.4	21.2	5:12	290	susp	-	-
G	3	602.9	34.0	8:12	179	susp	flag	-
H	1	776.2	33.1	8:12	358	susp	-	-
TOTAL		**3,559.3**				**see caveat**		

Label → segmentation index (cross-references the freeze and comparison files):

`A=F5    B=F4    C=F2    D=F6    E=F7    F=F0    G=F3    H=F1`

Total-level caveat (why the total is not a bare number):

the total inherits the dormer contamination, not just the facets: 6 of 8 facets are dormer-suspect, ~8 dormers are unmodeled with their points absorbed into host facets, and the resulting bias on the total is unquantified and not correctable by a flat allowance (decision 2026-07-15)

Flags: Dormer = per-facet dormer contamination (absorbed, unmodeled); susp* = heaviest. Eave 'flag' = wide loose/tight eave bracket (possible gutter/fascia/vegetation). Panel = solar region (later houses).

Areas per pitch, waste-to-squares, and the honest line-length inventory.

Area per pitch band

Band	Facets	Area ft²
Shallow (10-26 deg)	A/B/C/F	1,375.3
Steep (26-38 deg)	D/E/G/H	2,184.0

Roofing squares by waste allowance 1 square = 100 ft²

Waste	Area ft²	Squares
0%	3,559.3	35.59
10%	3,915.2	39.15
12%	3,986.4	39.86
15%	4,093.2	40.93
17%	4,164.4	41.64
20%	4,271.2	42.71
22%	4,342.3	43.42

Squares computed from the total, which carries the dormer caveat.

Line-length inventory

What the pipeline classifies TODAY: it derives and tape-validates exactly two lines. It has no general edge-classification stage, so no full ridge/hip/valley/rake/eave inventory is claimed.

Type	Line id	Length ft	Count	Status
Ridge	length:r6,7	34.33	1	tape-validated
Rake (slope span)	lines:r1,3-e1	22.68	1	tape-validated

What a full inventory would still require (not built; not fabricated here):

- Rake vs eave: derivable from each boundary edge's angle to horizontal (an eave is level; a rake climbs the slope).
- Ridge vs valley: needs the dihedral sign where two facets meet (ridge = convex/peak, valley = concave/gutter).
- Drip edge vs plane-plane line: needs to distinguish a facet's boundary edge from an interior plane-intersection line.
- All of the above needs a per-facet edge-enumeration + classifier stage the pipeline does not yet have; eave DIRECTIONS are derived today, but not full eave lengths.

The accuracy page competitors do not publish. Failure reported first.

1. Scale provenance

One tape length sets the multiplier: length:r6,7 taped 412.0 in over 10.184 cu = 40.4541 in/cu (3.37118 ft/cu). The freeze's recorded end-sensitivity (3.911 cu) is reported, not corrected away; its meaning is defined in the freeze context. GPS scale error at this multiplier: 2.75%.

2. Independent cross-check

A second, independent tape on lines:r1,3-e1 checks the multiplier: predicted 272.1 in vs taped 267.0 in = 1.92% (budget ±2.47%) -> PASS. Edge-definition sensitivity is shown, not resolved: against the with-gutter tape (270.0 in) the disagreement is 0.79%. Implied area uncertainty ~3.9%.

3. Run 1 reported first: the failure that drove the fix

Run 1 (preregistered-2026-07-15.json) FAILED its cross-check at -3.96% (predicted 256.4 in vs 267.0 in taped). Diagnosis: the frozen ridge extent jumped a void to a 97-point assignment-artifact island and read ~6% long. The fix was a cloud-side contiguity rule (cut the extent at any along-line void wider than 60x point spacing), which moved the primary span 10.808 -> 10.184 cu and left every other number bit-identical. Run 2 then passes at 1.92%. Decisive arithmetic: run 1 total 3160.6 ft² vs run 2 3559.3 ft² (12.6% apart); run 1's widened ±8.1% interval would NOT have contained run 2's corrected total. Widening instead of diagnosing would have quoted an interval that misses the value.

4. Per-facet pitch vs inclinometer

Pitch is validated directly against three inclinometer readings per facet. Truth uncertainty = the spread of those three; pipeline uncertainty = the 0.2° leveling floor. Max |error| 2.19°: PASS at 3° (True), not at 2° (False).

Facet	Pipe °	Truth °	Err °	Spread
A	21.4	19.3	+2.02	1
B	21.0	19.3	+1.66	1
C	22.7	20.7	+2.06	1
D	33.8	31.7 (90-x)	+2.09	1
E	33.5	32.0 (90-x)	+1.52	0
F	21.2	19.0	+2.19	2
G	34.0	32.0	+2.01	2
H	33.1	32.0	+1.07	0

5. The systematic pitch offset (OPEN)

Errors carry a uniform +1.83° offset. Residual cloud tilt is ruled out (the fitted azimuth term is not distinguishable from zero at this noise level). Named suspects, left unadjudicated: inclinometer zero/convention (the same instrument the F6/F7 90-minus reading flagged), and shingle-surface vs fitted-plane definition. Net of the offset, geometric scatter is ~0.24° rms.

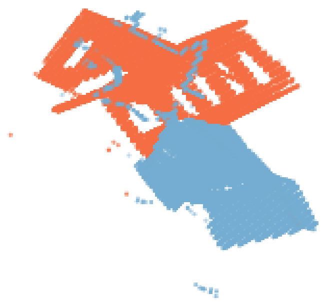
6. Claims

- Area scale is confirmed by an independent length.
- Area itself is NOT validated against a measured area, because none exists for this property.

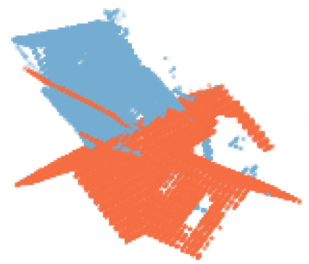
scale confirmed by an independent length (weaker claim); NO measured area exists for this property, so area itself is NOT validated against ground truth

Pitch-colored 3D, two elevations, and dormer-suspect annotation (individual dormers are unsegmented).

oblique azim 45

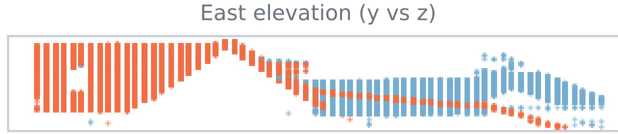


oblique azim 225



South elevation (x vs z)

East elevation (y vs z)



Dormer annotation

Dormer-suspect facets: C, D, E, F, G, H. Near-clean facets: A, B.

About 8 dormers were not segmented as their own facets; their points were absorbed into the host facets above, so the affected per-facet areas and pitches are biased in an amount that is not cleanly separable from foliage and an assignment artifact. Individual dormers therefore cannot be drawn; the honest annotation is which host facets are contaminated. Only the near-clean facets are safe to cite per-facet (decision 2026-07-15).